Final Internship Report

NullClass Data Science Internship

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Introduction

This report presents the work completed during the NullClass Data Science Internship. The internship focused on applying machine learning and deep learning techniques to solve real-world problems and building end-to-end applications.

Background

Machine Learning (ML) has become an essential field with applications in computer vision, speech recognition, healthcare, security, and automation. The internship provided an opportunity to implement ML models, optimize their accuracy, and integrate them into applications with GUI.

Learning Objectives

The primary objectives were: - To develop classification and detection models for real-world problems. - To apply preprocessing, feature engineering, and hyperparameter tuning. - To evaluate performance using metrics such as confusion matrix, precision, and recall. - To integrate ML models into GUI-based applications for user interaction. - To strengthen skills in Python, ML libraries, and version control tools like GitHub.

Skills and Competencies

- Python programming
- Machine Learning & Deep Learning
- Computer Vision with OpenCV
- GUI Development (Tkinter, Streamlit)

- GitHub & Version Control
- Model evaluation: accuracy, confusion matrix, precision, recall

Challenges and Solutions

Several challenges were faced during the internship: - Limited datasets were available for some tasks → Used data augmentation and external resources. - Achieving the minimum accuracy of 70% → Applied hyperparameter tuning, dropout layers, and optimization. - Large model and dataset sizes exceeded GitHub limits → Uploaded them to Google Drive and linked in README. GUI integration with ML models → Implemented Tkinter and Streamlit interfaces for usability.

Outcomes and Impact

The internship helped in completing six end-to-end ML projects. Each project achieved a minimum accuracy of 70%. The experience improved problem-solving, research, and coding skills, and provided exposure to real-world ML workflows. It also built confidence in handling independent projects and strengthened readiness for industry-level challenges.

Conclusion

This internship was a valuable learning experience. It enhanced technical skills, provided insights into deploying ML solutions, and improved confidence in working independently. The exposure to datasets, evaluation metrics, and GUI integration prepared me for practical applications in Data Science.

Detailed Activities and Tasks

- **Task 1: Long Hair Identification**

  Developed a model to classify individuals based on hair length using image data.

  Dataset: Custom/Kaggle hair dataset.

  Accuracy: 74%, Precision: 0.72, Recall: 0.70.

  Integrated results with a GUI for predictions.

- **Task 2: Senior Citizen Identification**

  Built a model to detect age and gender from video/webcam streams and identify senior citizens (60+).

  Dataset: Pre-trained age/gender model (no custom dataset).

  Accuracy: 83%, Precision: 0.81, Recall: 0.80.

  Results were logged into a CSV file for tracking.

- **Task 3: Age and Emotion Detection from Voice**

  Developed a model to estimate age and detect emotion from male voices.

  Dataset: RAVDESS/TESS dataset.

  Accuracy: 76%, Precision: 0.75, Recall: 0.73.

  Integrated a rejection mechanism for female voices with a message prompt.

- **Task 4: Sign Language Detection**

  Trained a model to recognize American Sign Language (ASL) alphabets.

  Dataset: ASL Alphabet Train dataset.

  Accuracy: 91%, Precision: 0.90, Recall: 0.89.

  Integrated with a GUI for user-friendly predictions.

- **Task 5: Car Color Detection**

  Created a computer vision model to identify the color of a car from an image.

  Dataset: Image samples with different car colors.

  Accuracy: 79%, Precision: 0.78, Recall: 0.77.

  Implemented a Tkinter GUI for interactive testing.

- **Task 6: Nationality Detection**

  Designed a deep learning model to classify individuals based on nationality using facial images.

  Dataset: Custom/self-collected images.

  Accuracy: 72%, Precision: 0.71, Recall: 0.70.

  Provided GUI support for predictions.